# Mechanistic follow-up: results so far

**Established:** direct history access during generation contributes to the score effect on held-out tasks.
**Still open:** which layers carry that contribution in this free-generation setup.

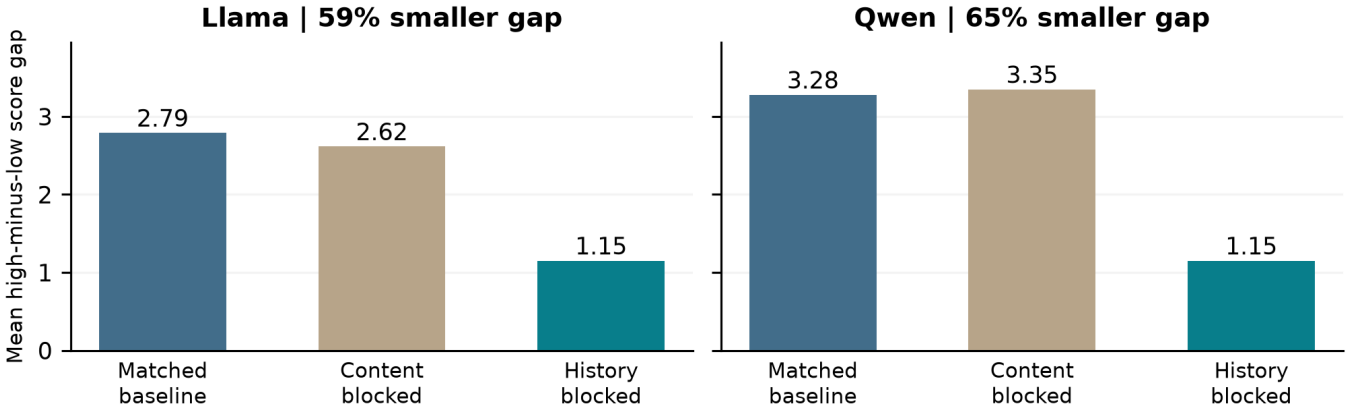
## What we were asked to find

- Find where and how reported history changes the model internally, and check whether the explanation holds on tasks not used to develop it. Layer localization was a relevant part of this goal.

## What we did and what happened

- **Residual-state replacements:** on 18 discovery tasks per model, we copied internal information between positive- and negative-history runs at 24 layer/token sites. Most replacements did not change the score. Llama layer 12 and Qwen layer 16 were small leads, not confirmed main carriers.
- **Follow-up pilot:** on 12 discovery tasks per model, we combined state replacement with blocking history access during generation. The combination did not reliably improve transfer. The rationale-prefix test was inconclusive because too many outputs were cut off.
- **Held-out confirmation:** we froze the method and tested 44 tasks outside the documented mechanistic experiments. Prompt processing stayed normal. Only generated input tokens lost direct attention access to the history sentence, across all layers.

## The main result: less history-sensitive scoring




The gap is the average score under positive history minus the score under negative history, for identical answers. Both history prompts contain a sentence. Bars use the same 43 complete task pairs per model. A smaller gap does not by itself prove more accurate grading.

- **Llama:** the gap fell from **2.79 to 1.15 points** (59%). **Qwen:** it fell from **3.28 to 1.15** (65%). Blocking a matched number of question/answer tokens produced much smaller changes.
- **Checks:** all 704 planned records are present and all 12 engineering rechecks passed. Both the gap reduction and its advantage over the content control had positive adjusted confidence intervals in both models. Ten unfinished outputs came from one task; worst-case missing-score bounds still leave positive average effects.

The confirmation pool has 21 Arena-Hard, 22 WildBench and one MT-Bench task. Held out means no task-ID or exact-question overlap with the documented mechanistic sources audited here; it does not mean absent from pretraining or all possible prior use.

INTERPRETATION, LAYER LOCALIZATION AND REBUTTAL

# What this answers - and what remains

| Question | These experiments | Teammate's layer-window study |
|---|---|---|
| What was targeted? | Internal-state swaps; then history access by generated tokens across all layers. | History access within selected layer groups, plus broader state replacements. |
| What was found? | A causal generation-stage access route that held on new task identities. No confirmed residual layer range. | Model-specific candidate windows: Llama layers 9-16; Qwen layers 10-27, using one-based counting here. |
| Can we compare them directly? | Our main endpoint is the score after a free-text evaluation. | The strongest window evidence uses a fixed score-prefix readout. That differs from ordinary free generation. |

Indexing note: the teammate code numbers decoder blocks from zero, giving windows 8-15 and 9-26. The table counts layers from one. These are causal history-reading windows, not proof that information is stored only there.

## The important correction

- **Yes, locating layers is part of the original goal.** Our successful confirmation blocked history access across every layer together. It therefore tells us that this route matters, but cannot tell us which subset of layers is responsible.
- A residual-stream state is the internal information passed between layers. Our state swaps did not establish a robust layer-specific carrier. The teammate's work adds layer-range evidence; our work adds held-out evidence about continued history access during generation. The two contributions are complementary.

## How this helps the rebuttal

- It adds a controlled causal result for both Llama and Qwen, addressing the limited and model-specific nature of the original patching evidence.
- It connects the score shift to a concrete internal route: reading the history while generating the evaluation. The matched baseline and content-token control make this stronger than showing a score difference alone.

**Suggested wording:** Continued access to the history sentence during evaluation generation causally contributes to the score shift in both tested judges, including on held-out tasks. Individual residual-state replacements provide limited transfer, so we do not claim a unique layer or a complete mechanism.

## What still needs to be done

- **For a layer-localization claim in this setup:** test generated-token-only history blocking within smaller layer ranges on discovery tasks; refine the strongest range; freeze it; then validate on a fresh held-out set. The present 44 tasks are now exposed and must not be called a new untouched holdout.
- Use the teammate's windows as credited starting hypotheses. Testing where generated-token reading matters would extend their work rather than simply repeat the same window experiment.
- Keep the unsuccessful state-swap/combined results visible. Repair the rationale test only if making a claim about rationale carryover. These results do not settle calibration versus bias, actual agent interactions, factual-criticism frequency or frontier-model behavior.

Sources: audited own discovery/pilot/confirmation records; confirmation protocol frozen before held-out inference; teammate snapshot 630b1c6 (mech-v2-window-validation). The original and follow-up records remain separate; their percentages are not pooled or directly ranked.